

Online Retail Customer Behavior Analysis

1. Project Overview

This project analyzes customer purchasing behavior for a UK-based online gift-ware retailer, using real transactional data covering 541,910 line items recorded between December 2010 and December 2011. The goal is to uncover which products and markets drive revenue, understand customer loyalty and value, and quantify the business impact of order cancellations and unidentified (guest/wholesale) transactions, in order to guide retention, marketing, and inventory decisions.

2. Dataset Summary

- **Source:** Online Retail II (UCI Machine Learning Repository)
- **Raw rows:** 541,910
- **Rows after cleaning:** 531,392
- **Columns:** 8 — Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country
- **Date range:** December 2010 – December 2011
- **Countries represented:** 38
- **Unique identified customers:** 4,372
- **Unique products:** ~4,070
- **Missing data (raw):** Description — 1,454 missing (0.27%); Customer ID — 135,080 missing (24.9%)

3. Data Preparation using Python

Data preparation and cleaning was done in Python using pandas:

- **Data Loading:** Imported the dataset using **pandas**
- **Initial Exploration:** Used **df.info()** to check structure and **.describe()** for summary statistics.

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
count	541910	541910	540456	541910.000000	541910	541910.000000	406830.000000	541910
unique	25900	4070	4223	NaN	23260	NaN	NaN	38
top	573585	85123A	WHITE HANGING HEART T-LIGHT HOLDER	NaN	10/31/2011 14:41	NaN	NaN	United Kingdom
freq	1114	2313	2369	NaN	1114	NaN	NaN	495478
mean	NaN	NaN	NaN	9.552234	NaN	4.611138	15287.684160	NaN
std	NaN	NaN	NaN	218.080957	NaN	96.759765	1713.603074	NaN
min	NaN	NaN	NaN	-80995.000000	NaN	-11062.060000	12346.000000	NaN
25%	NaN	NaN	NaN	1.000000	NaN	1.250000	13953.000000	NaN
50%	NaN	NaN	NaN	3.000000	NaN	2.080000	15152.000000	NaN
75%	NaN	NaN	NaN	10.000000	NaN	4.130000	16791.000000	NaN
max	NaN	NaN	NaN	80995.000000	NaN	38970.000000	18287.000000	NaN

- **Missing Description Handling:** Backfilled missing product descriptions using a StockCode-to-Description lookup built from other rows sharing the same product code — recovered 1,342 of 1,454 missing values; the remaining 112 rows were dropped.
- **Missing Customer ID Handling:** Deliberately not dropped. Missing Customer ID likely represents guest or wholesale checkouts rather than bad data, so these rows were kept for revenue/product-level analysis and only excluded from customer-level analysis (RFM, loyalty segmentation) at the SQL layer.
- **Duplicate Removal:** Removed 5,268 fully duplicate rows.

- **Non-Product Row Removal:** Removed 2,754 rows carrying non-product StockCodes (POST, DOT, M, D, S, AMAZONFEE, CRUK, PADS, B, and similar) representing postage, fees, and manual adjustments.
- **Invalid Price Removal:** Removed 2,384 rows with Price ≤ 0 .
- **Country Standardization:** Replaced "Unspecified" and "European Community" with a single "Unknown" label.
- **Cancellations and Returns:** Deliberately not removed — kept to measure cancellation rate and lost revenue as a real business finding.
- **Database Integration:** Loaded the cleaned dataset into PostgreSQL; business-logic labeling (customer segment, cancellation flag) was built as SQL views rather than in Python.

4. Data Analysis using SQL

Structured analysis was performed in PostgreSQL to answer 10 business questions:

1. Revenue by Country – Compared total revenue generated across the top markets.

	Country character varying (50) 🔒	total_revenue numeric 🔒
1	United Kingdom	8726474.50
2	Netherlands	283889.34
3	EIRE	276090.86
4	Germany	205381.15
5	France	184679.00
6	Australia	138103.81
7	Spain	55706.56
8	Switzerland	53065.60
9	Japan	37416.37
10	Belgium	36927.34
11	Sweden	36828.83
12	Norway	32454.64
13	Portugal	26951.11
14	Channel Islands	20147.54
15	Finland	18344.88
16	Denmark	18211.34
17	Italy	15820.24
18	Cyprus	13167.16
19	Hong Kong	9744.19
20	Singapore	9120.39
21	Austria	8742.68
Total rows: 37		Query complete 00:00:00.294

2. High-Spending Returning Customers – Identified customers who had at least one returned order but still spent above the average customer total.

	Customer ID numeric	total_spent numeric
1	14646.0	278778.02
2	17450.0	189575.53
3	14911.0	132893.24
4	12415.0	123638.18
5	14156.0	114335.77
6	17511.0	88138.20
7	16684.0	65920.12
8	13694.0	62961.54
9	16029.0	60369.93
10	15311.0	59284.19
11	13089.0	57322.13
12	15061.0	54250.34
13	17949.0	53215.74
Total rows: 623		Query complete 00:00:00.723

3. Top 5 Products by Revenue – Found the highest-revenue-generating products.

	StockCode character varying (20)	Description character varying (255)	total_revenue numeric
1	22423	REGENCY CAKESTAND 3 TIER	174156.54
2	23843	PAPER CRAFT , LITTLE BIRDIE	168469.60
3	85123A	WHITE HANGING HEART T-LIGHT HOLD...	104284.24
4	47566	PARTY BUNTING	99445.23
5	85099B	JUMBO BAG RED RETROSPOT	94159.81

4. UK vs. Non-UK Order Value – Compared average order value between UK and international customers.

	market text	avg_order_value numeric	num_orders bigint
1	UK	394.74	20966
2	Non-UK	659.36	2268

5. Identified vs. Unidentified Customers – Compared average order value between logged-in and guest/wholesale transactions.

	customer_segment text	avg_order_value numeric	num_orders bigint
1	Identified	379.48	21799
2	Unidentified	1044.77	1435

6. Highest Cancellation-Rate Products – Identified 5 products with the highest cancellation rate.

	StockCode character varying (20)	Description character varying (255)	cancelled_count bigint	total_count bigint	cancellation_rate_pct numeric
1	BANK CHARGES	Bank Charges	25	37	67.57
2	23064	CINDERELLA CHANDELIER	10	33	30.30
3	22656	VINTAGE BLUE KITCHEN CABIN...	6	26	23.08
4	23462	ROCOCO WALL MIRROR WHITE	12	57	21.05
5	22461	SAVOY ART DECO CLOCK	13	66	19.70

7. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on number of distinct orders placed.

	customer_segment text	num_customers bigint
1	Returning	1969
2	New	1325
3	Loyal	1068

8. Top 3 Products per Country – Listed the most purchased products within each country by quantity sold.

	Country character varying (50)	StockCode character varying (20)	Description character varying (255)	total_qty bigint
1	Australia	22492	MINI PAINT SET VINTAGE	2952
2	Australia	23084	RABBIT NIGHT LIGHT	1884
3	Australia	21915	RED HARMONICA IN BOX	1704
4	Austria	21918	SET 12 KIDS COLOUR CHALK STICKS	288
5	Austria	22546	MINI JIGSAW PURDEY	240
6	Austria	22584	PACK OF 6 PANNETONE GIFT BOXES	144
7	Bahrain	23076	ICE CREAM SUNDAE LIP GLOSS	96
8	Bahrain	72802B	OCEAN SCENT CANDLE IN JEWELLED BOX	60
9	Bahrain	23077	DOUGHNUT LIP GLOSS	60
10	Belgium	21212	PACK OF 72 RETROSPOT CAKE CASES	480
11	Belgium	22326	ROUND SNACK BOXES SET OF4 WOODLA...	420
12	Belgium	22417	PACK OF 60 SPACEBOY CAKE CASES	384
Total rows: 108 Query complete 00:00:00.469				

9. Repeat Buyers and Spend – Checked whether customers with more than 5 orders spend more on average than non-repeat buyers.

	buyer_type text	num_customers bigint	avg_total_spent numeric
1	Non-Repeat Buy...	3294	693.53
2	Repeat Buyer	1068	5606.54

10. Revenue by Month – Calculated total revenue contribution of each month, to identify seasonal patterns.

	year_month text	total_revenue numeric
1	2010-12	776329.95
2	2011-01	670654.46
3	2011-02	508065.09
4	2011-03	690466.82
5	2011-04	515878.08
6	2011-05	740451.14
7	2011-06	738233.99
8	2011-07	688728.12
9	2011-08	724688.42
10	2011-09	1029268.80
11	2011-10	1104057.63
12	2011-11	1453292.69
13	2011-12	614710.96

5. Dashboard in Power BI

An interactive dashboard was built in Power BI to present these insights visually, including:

- KPI cards: Total Revenue, Number of Customers, Average Order Value, Cancellation Rate %
- Donut chart: Share of orders from identified vs. unidentified customers
- Bar chart: Customer segment distribution (New / Returning / Loyal)
- Bar chart: Revenue by country (top 10)
- Line chart: Revenue by month (Dec 2010 – Dec 2011), showing seasonal trends
- Bar chart: Top 10 products by quantity sold

Customer Behavior Dashboard

Customers Subscription

Identified

Unidentified

Country

Australia

Austria

Bahrain

Belgium

Brazil

Months of the Year

2010-12

2011-01

2011-02

400K

Number of Customers

441.37

Average Order Value

14.84

Cancellation Rate %

\$10.25M

Total Revenue

% of Identified vs Unidentified customers



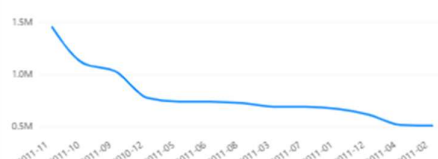
Customer Segment counts



Revenue by Country



Revenue by Month



Top 10 Products by Quantity

